National Institute of Business Management School

Electronic Health Records Quality Scorecard

Higher National Diploma in Software Engineering

Machine Learning

Submitted By

COHNDSE251F-041	W D Y N Gunawardana
COHNDSE251F-042	W A H Ishan
COHNDSE251F-058	W W Thilini Piyumika
COHNDSE251F-059	P A R Sahas Suraweera

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Abstract

Electronic Health Records (EHR) are widely utilized in the healthcare industry for research and analytics in hospitals and related institutes. Those records often contain quality issues like missing values, inconsistencies, incomplete records, duplications etc. However, when it comes to decision making and drawing conclusions, those issues can result in bias and inaccurate results. In further, it is reasonable for low performance in machine learning models which are being created for the required purposes. To address this issue by overcoming those challenges, we propose our solution which is a machine learning based approach which enables the evaluation of EHR before using it for analysis and the model development. It evaluates the research readiness of the EHR to ensure the completeness and data quality. It deployed as a pre analysis screening tool for healthcare researchers, professionals and other stakeholders and it will reduce bias, enhance data and accuracy of AI and ML models which utilize Electronic Health Records.

Keywords - Electronic Health Records (EHR), Machine Learning, Healthcare, Research and Development, Research Readiness

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1. Introduction

1.1 Background and Usage of Electronic Health Records

In healthcare industry, clinical documentations are drawn using Electronic Health Records (EHR). Usually, they are being utilized beyond their intended purpose (original purpose) such as in healthcare analytics, clinical research, systems for decision support, and for developing models implemented with artificial intelligence and machine learning. Those records can store thousands, or even millions of data allowing researchers to identify anomalies or patterns of diseases and outbreaks (Buntin et al., 2011).

Hence, large-scale health studies are vitally depending on these records, and they aggregate many crucial features. Longitudinal nature, large in volume and featuring and covering the real-world scenarios can be mentioned as some of them. Despite their valuable impact, quality issues of EHR datasets matters the most. It happens because they were not originally designed and created to for the research and analytical purposes. Low quality data across various health care settings would create a negative impact on many factors such as fairness in results, validity and reproducibility of downstream analyses and machine learning models. As a matter of fact, biased or misleading outcomes would be appeared if the necessary quality is not maintained by not achieving the minimum data standards prior applying the AI techniques. Our research-oriented solution has overcome those challenges, and through a structured and interpretable framework model will validate the research readiness of EHR data as it satisfies the industrial needs of the stakeholders.

1.2 Problem Statement

Utilization of Electronic Health Records is being comparatively increased during the past decade in hospitals and institutes in United States (ONC, 2023) and around the world and they are being actively executed for research purposes as they mirror the real-world clinical data. To take healthcare decisions and building software solutions to solve the problems regarding healthcare systems these records are essential. But challenges like missing information, incomplete documentations and duplicate records can cause bias in analysis, incorrect conclusions and provide imbalanced data for decision making in healthcare sector.

According to the Gianfrancesco et al. (2018) issues in Electronic Health Records can significantly degrade the machine learning model performance. That study highlights missing data in EHR is the main reason for the biasness and misclassification which leads to poor conclusions. In USA, major incident had happened related to EHR. A company had developed an Epic Sepsis prediction algorithm, who utilized EHR to develop and implement their model. Due to the low quality of data model fails and the hospitals reported false alarms, missed cases and poor performance (Ross, 2021). Since the intended requirements expect accurate and precise outputs while the decisions and conclusions are highly sensitive for human, developing a model to classify quality EHR related data is important and required by the stakeholders.

1.3 Objectives

- Extracting features which are meaningful from raw EHR data.
- Analysing the significance of different factors during the documentation which complements EHR data quality.
- Developing a model that which classify and mirror the documentation completeness of EHR records for research purposes and decision making.
- Building structured and flexible dataset by transforming heterogeneous EHR.
- Evaluating the capability of differentiating reliable and unreliable EHR data.
- Deploying the model as a pre-analysis screening tool for healthcare professionals, researchers and other stakeholders.

2. Literature Review

2.1 Introduction

Electronic health records are for several purposes other than clinical recordkeeping. Those data use for clinical research, healthcare analytics, healthcare decision support and for the building healthcare related artificial intelligence models. However, Electronic health record systems are not initially developed for research, leading to documented data quality issues that threaten the validity of future analyses. This review provides key scientific evidence suggesting the existence of a gap in research ready EHR data and need of quality structured rating frameworks and explaining importance of using machine learning to build a solution for reusability of valid EHR data.

2.2 Use of EHR Data in Research

Recent research shows that EHRs are mainly for clinical admirative purposes and documentation and mainly not designed for research , however those data is more helpful and valuable because of its large amount but there is a significant complexity using that , there are several studies proposed solution for make EHR data research ready using three grouped categories first one is data curation and standardization, machine learning and advanced statistical methods and frameworks.

2.3 Existing Approaches

EHRs are designed and developed for clinical and administrative use especially for patients' documentation not for directly use for research. This creates a challenges and limitations for biomedical and health service research when they are using an electronic health record. Despite these barriers, the large amount, continuous EHR data provides a significant opportunity for research and large-scale healthcare observations and other health analysis (Singhal et al., 2023). According to Sauer et al. (2022), there are some limitations in EHR data and EHR data are collected during regular clinical practice, and it is resulting in inconsistency in data quality and standards of the data. Getzen et al. (2023) introduced an interesting method for replicate missing data they have found the innovative solution for key problem in EHR data that is missing values. Those missing data are not random, and this can be led to build bias machine learning models. They have defined EHR data foe three main levels which is level 0 which is a raw EHR data without processing, level 1 which is structured and unstructured and level 2 matrix style processed data. They have developed framework which can simulate missing data based on 3 missingness, random missingness, missingness on observed patient characteristics, missingness due to unseen conditions. Their research developed a modelling missing EHR data using knowledge graph and shows how significantly affects ML models and disadvantaged patient populations. According to Callahan et al. (2020) EHR data is valuable but not research ready and they are pointing out that there is a standard guidelines like STROBE and RECORD however, EHR data require further considerations.

This research shows that time stamp is not a detail that can reflect that event happened and notes could be contain further information and that could lead to data leakage. Their finding shows just using a machine learning model is not enough and it should assess the quality of data.

2.3 Gaps in Existing Work

Many existing methods and frameworks always tried to improve medical data by just cleaning and adjust to fit and, they always focused on improving model performance rather than reliability of actual 2 EHR data. Most of the models check the data quality however it fails to show which data is quality enough to use for research. Also, differences in clinical systems patients' group are ignored most of the times and that makes harder to decide whether these data are reliable or not for the analysis.

2.4 Proposed Solution

To fill these gaps, we propose an interpretable EHR Data Quality Scorecard and classification model that can use for research, clinical analysis and healthcare decision. Model uses key documentational factors such as completeness, conflicts, temporal consistency, and documentation modality to assess if an EHR record is research ready or not. Rather than eliminating bias after building the modelling, our technique allows for early detection of incorrect and biased records, resulting in more transparent, fair, and efficient EHR-based research.

3. Methodology

3.1 Problem Definition

As first step in machine learning lifecycle, it is important to identify the problem is a binary classification task. When it comes to electronic health records, indicate those records are reliable on research or not is our main objective. Hence, it should clearly define whether those data are research ready or not. This problem formulated as a classification task rather than clustering since the binary outcome is expected. Hence the supervise learning techniques are selected, evaluated, compared and implemented in further steps of the project.

3.2 Data Collection

Most Electronic health records are under privacy and ethical restrictions. However, open-source Electronic Health record dataset is selected for the project which is available on publicly for research and educational purposes (See Appendix for source information). Dataset was a realistic EHR dataset which changed patients' names, addresses replaced to ensure privacy. The original data set was presented as multiple tables which is included by patients, encounters, conditions, observations, procedures etc. Once after data tables ready then transformed into a training dataset. Initially, each table was examined to understand its schema and the common patient identifier used to link records. The tables were then made into one excel sheet by joining them through the patient ID, and relevant documentation signals were used to create meaningful patient-level features. cleaning missing values, standardizing the structure, and converting multiple tables into a single complete dataset where each row represents one patient record. The resulting table ensure efficient feature extraction and supported the model training.

3.3 Data Preprocessing

As for the first step excel sheet which contain the dataset was imported to a 'pandas' data frame and then handled missing values, after that applied feature engineering techniques to identify what are the existing features and then created meaningful features using those features that supported model to learn patterns in a better way. Rather than rely on raw counts used ratio-based features for an example observations per visit and medications per condition used to identify documentation density. Some features contain very large values compare to other values this value difference cause to imbalance, hence applied log transformation to reduce extreme differences of values.

Log transformation Formula,

$$X' = \log(1 + X)$$

X' = Transforming the value of X

X = Original value

log = Logarithm function (base e)

Used log(1+X) because log (0) is undefined. This formula makes larger numbers smaller. StandardScaler was applied for Logistic Regression to normalize features, standardized data could help to Logistic Regression perform better.

$$Z = \frac{X - \mu}{SD}$$

$$\begin{aligned} Z &= \text{Standardized value of } X \\ X &= \text{Original value} \\ \mu &= \text{Mean of } X \\ SD &= \text{Standard deviation of } X \end{aligned}$$

Standardization helps domination of features contain higher values. And help to treat all features fairly. Overall, those techniques enhanced model performance and stability. As a next step target variable (Y)and features (X)are divided. Removed Patient ID and Richness score to prevent data leakage because if model can identify those columns, it can directly predict the answer. Unique identifies like patient ID can cause data leakage because it contains random numbers and string values. In this model built a richness score using other features and then used richness score to create readiness score also richness score is a derived variable, and model can do reverse engineering and get prediction easily, and in machine learning it is called target leakage. And for the model evaluation split the dataset into two parts which is testing dataset and training dataset. 70% for the training dataset and 30% for the training dataset ensure stratified sampling to maintain class distribution. Data preprocessing was helpful to identify data leakages and understand how stable model training process.

3.4 Model Selection

When selecting a model focused on three classification models. First one is logistic regression, second one is decision tree and last one is random forest classification model. These three models were selected to evaluate which model is best fitting and understand the different approaches of each model and how accuracy could be change based on the model. As a basic model chose a logistic regression to evaluate, if data can sperate in a linear way it helps to understand the problem is complex or simple. Decision tree was used because it can take decisions step by step like a flow chart then can identify and non-linear relationships. Then used random forest because it is a combination of many decision trees and help to make predictions more accurate.

3.5 Model Training

Each model was trained using previously split training dataset each model learns patters and relationships between input features and target variable. In Logistic regression it allocates weights to each feature. If weight is too large, then model will overfit. Because of that used L2 regularization to avoid large weights it adds penalty for large weights

L2 Regularization formula,

$$Loss = \text{Original Loss} + \lambda \Sigma w^2$$

$$\begin{aligned} \textbf{Loss} &= \text{Total loss function} \\ \textbf{Original Loss} &= \text{Loss without regularization penalty} \\ \lambda\ (\text{lambda}) &= \text{Regularization strength} \\ \Sigma w^2 &= \text{Sum of the squares of the model weights} \end{aligned}$$

Using this formula forced to weights to stay smaller not to become larger. In regularization we can control it by using C value, if C value is small that means strong regularization and if C value is large that means weak regularization based on that used C = 0.5 as it represents moderate regularization. When it comes to decision tree depth of the decision tree was controlled because if tree grows too deep it will try to memorise data and will become complex as well. When it comes to random forest build it using 300 decision tress and each tree contains

$$R' = R + \epsilon$$

$$\begin{aligned} R' &= \text{Adjusted or noisy richness score} \\ R &= \text{Original richness score} \\ \epsilon &= \text{Gaussian noise} \\ \Sigma w^2 &= \text{Mean} = 0,\ \text{SD}\ (\sigma \doteq \text{spread}) \\ & \quad \text{standard deviation of the noise} \end{aligned}$$

max_depth =4, in random forest many trees can maker stable predictions. Then build research richness score by combining features such as visits, observations, conditions, medications and procedures. Controlled Gaussian noise was added to the richness score using below mentioned formula,

This formula makes a new score by adding random noise to the original score.

$$\epsilon \sim N(0,\sigma)$$

Random number comes from normal distribution with Mean =0 and Standard deviation = σ

main purpose of using this noise is to avoid perfectly learn the threshold rule. If that happens model will give high unrealistic accuracy. This noise adding help model learn true patterns.

3.6 Model Evaluation

Performance of each model were evaluated by using accuracy, precision, recall and F1-score. These measurements are good enough to understand overall performance of the model. Accuracy indicates the how many predictions are correct, precision checks how often the predictions are correct and avoid false positivity. Recall checks how many positive cases model found correctly, especially in healthcare data missing poor quality data is matters. F1 score gives a balanced value using precision and recall values. For the evaluation used test data then can check whether model perform well on test data or not. Also performed cross-validation, in cross-validation instead of splitting data once split into 5 different combinations with 5 iterations, in each iteration it will train 4 parts and test on 1 part. Cross validation checks model stability across different data splits. To calculate cross-validation first get Mean Accuracy, then calculate standard deviation using mean accuracy then can identify how much model performance change in each iteration if standard deviation is small that means model is stable and if standard deviation is large that means model is unstable, then it requires more optimization.

$$\bar{A} = \frac{A_1 + A_2 + \cdots + Ak}{k}$$

$\bar{A}$ = Mean accuracy

$$\sigma = \sqrt{\frac{1}{k} \sum_{i=1}^{k} \bar{A_i}(A_1 - \bar{A})^2}$$

σ = Standard deviation

A_i = Accuracy in fold i

k = Number of folds

3.7 Tuning & Optimization

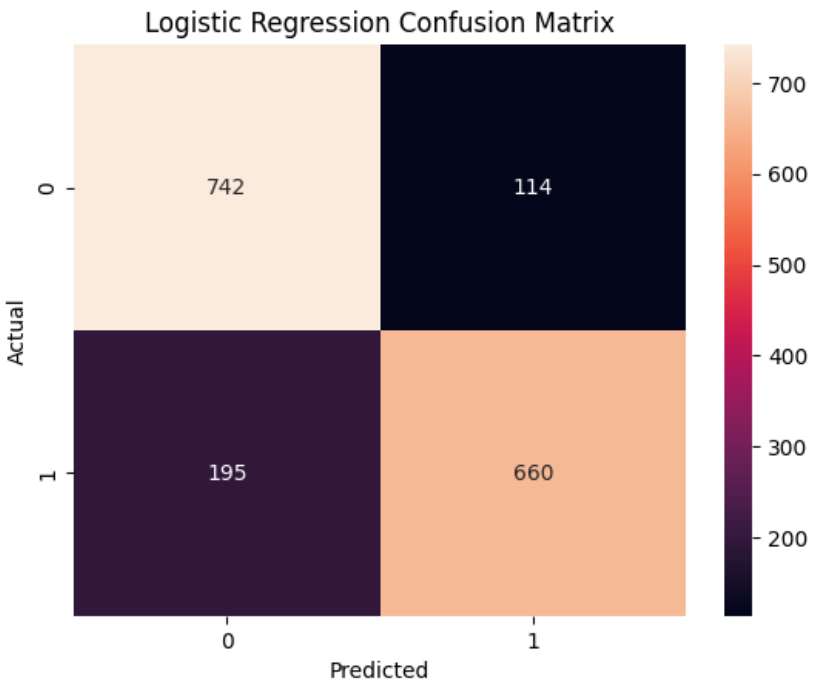
To optimize the model hyperparameters like tree depth, regularization strength and numbers of trees used in random forest were adjusted to enhance the accuracy and model stability, reduce overfitting and improve model stability, adjusting hyperparameters are important to avoid underfitting and overfitting. Used cross-validation and according to that model was optimized. Logistic Regression got the highest accuracy in cross validation which is 80.37%. this indicate that reduced overfitting and stable performance. Hence, logistic regression chose as the final model.

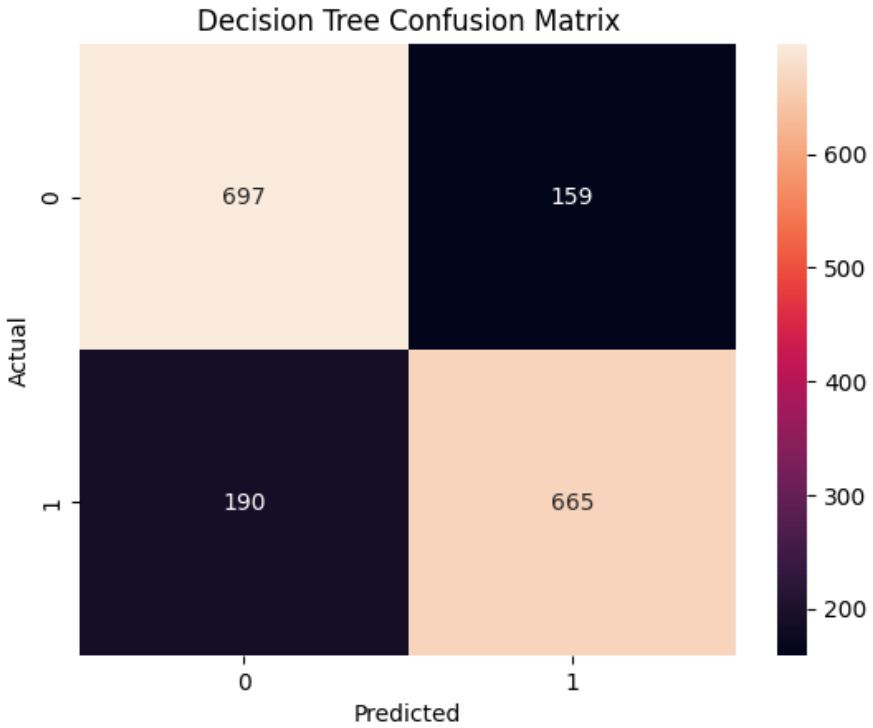
4. Results & Analysis

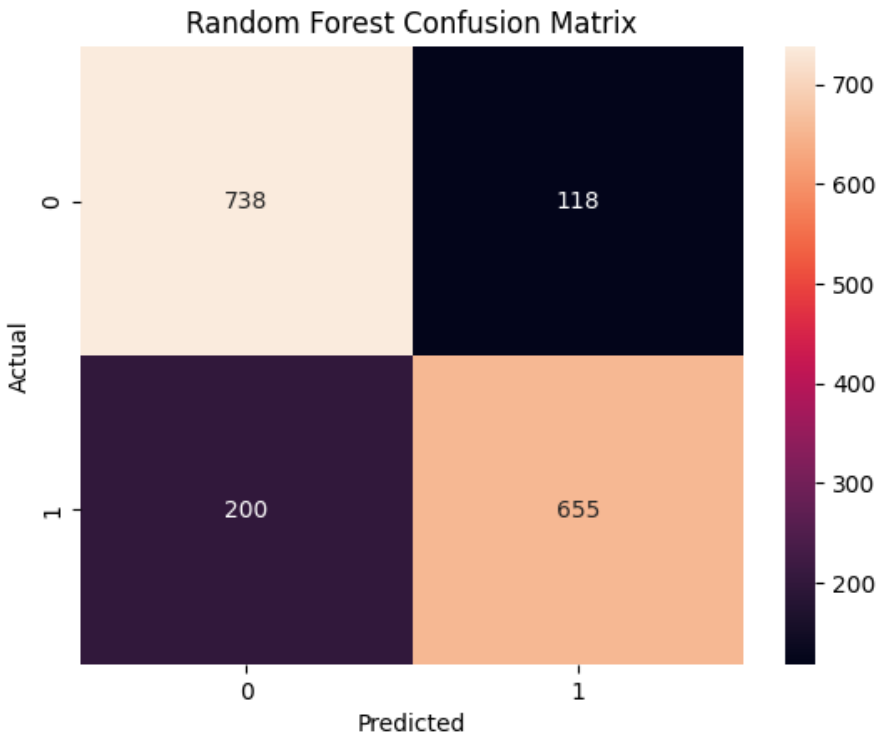
4.1 Model Performance

Making a comparison between each model accuracy rates and F1 scores. Random forest accuracy rate was high while logistic regression performance was stable. Decision tree showed higher accuracy, but random forest approach was provided reliable and accurate predictions.

4.2 Confusion Matrix Analysis



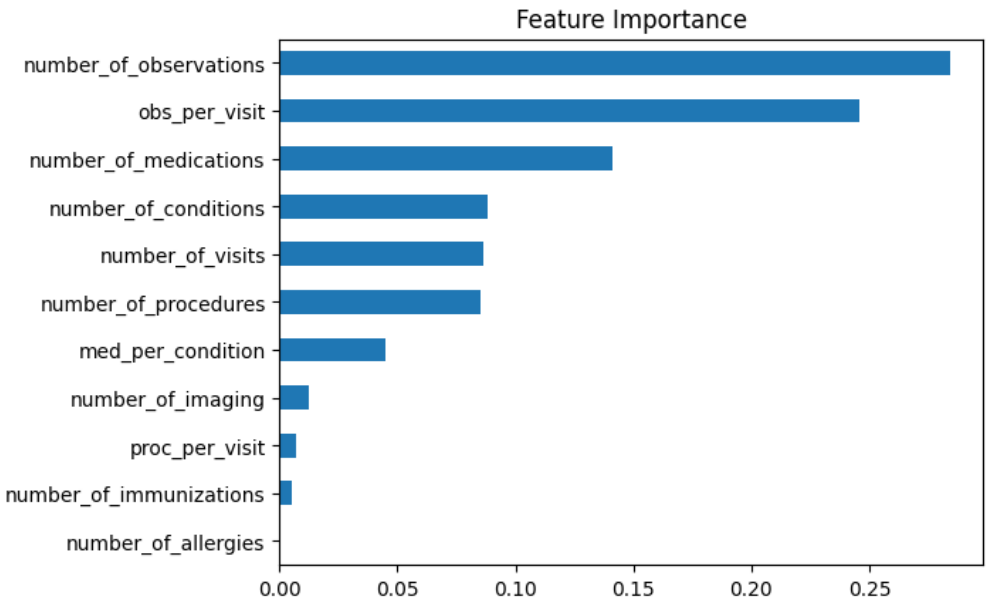




4.3 Feature Importance Analysis

When it comes to feature analysis, random forest indicated that some features are important while some are not. It demonstrated that all features ae not equally important to make predictions and evaluate the quality score. It provided a proper understanding about which features are most valuable to evaluate reliability make model more understandable. After the feature analysis results identified prioritized data which is helpful to make guidelines to protect consistency and data quality in EHR.

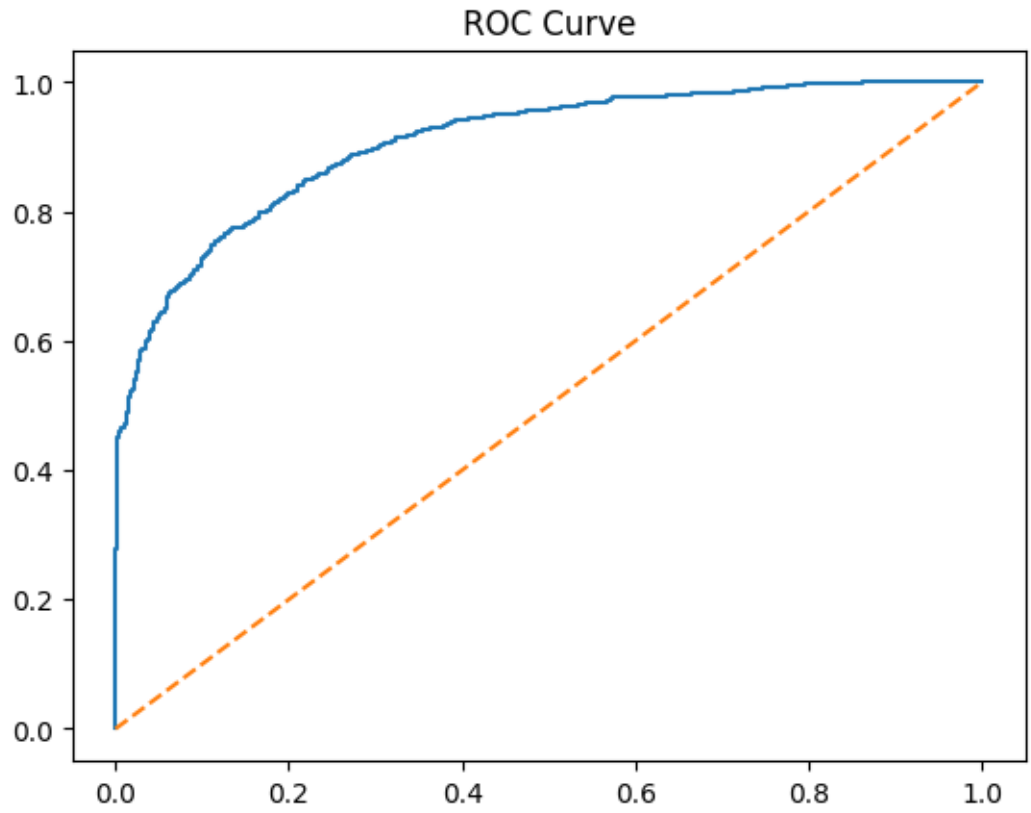
Feature Importance Bar Chart



4.4 Error Analysis

In borderline cases misclassification occurred which means some records show both reliable and not reliable feature values. Some feature values were overlapping between classes which means there's no significant differences between both classes. For an example, if reliable data records contain number of observations between 0-6 then unreliable record has number of observations between 2-10 between 2-6 there is an overlap to avoid this situation features do not create cleaner boundary. Also, some features did not provide enough information. Analysing feature engineering process was conducted by removing and adjusting new features and transforming variables. According to the feature engineering better features help separate classes and better separation helps reduce misclassification.

ROC (Receiver Operating Characteristic) Curve -Logistic Regression



In this ROC curve x-axis represent False Positivity Rate, which means among actual negatives how many models identified wrongly as positively-axis represent true positive rate which means out of actual positive values how many correctly identified. Lower x-axis and higher y-axis are identified as better. In this graph Orange dashed line represent random guessing if model is random, result will lie on that line. Blue steep curve shows that the model has high recall and false positivity relatively low. This graph assure that this model can clearly identify research-ready records

5. Discussion

This study is executed and demonstrated using several and relevant machine learning algorithms which are logistic regression, decision tree and random forest. Logistic regression had the highest cross-validation accuracy and results shows that machine learning techniques can be utilized to classify, validate and evaluate the research readiness of electronic health records by using structured documentation features. After training and testing more than three times identified model shows an unrealistic accuracy rate and near perfect linear separability. Then, Approach was changed by using a median threshold instead of a fixed threshold and added gaussian noise, regularization and cross-validation to get the accurate results.

Utilized dataset does not contained a predefined research readiness which should be the target variable(Y) in supervised learning. As mentioned in the problem definition expected outcome is to get a binary result hence followed supervised learning. To create a target variable and build a proxy indicator. Proxy indicator means alternative measurement for something that is unable to observe directly. When building this proxy indicator, we considered EHR data quality principles such as, documentation completeness and data density. According to Weiskopf & Weng (2013) Analysis of previous studies regarding healthcare informatics shows that consistency, completeness and documentation coverage are important and critical factors matters for research usability. Followed an established that past study that discussed research principles to construct a label that can indicate data reliability. One of the most important parts in this model training is instead of relying on separate numbers, it calculated a one total score as richness score by giving weight according to clinical relevance and documentation priority. After constructing richness score then focused on binary output which is research ready or not. To get this binary output used median by arranging smallest to largest also known as 50th percentile which is explaining 50% records are below it and 50% records are above median. Used median to get balance distribution then model will perform better and prevent bias. The Model calculates a probability first and then using a threshold rule it will give a binary outcome.

Through to the cross-validation results it is concluded that logistic regression was stable and gave accurate predictions. That indicates strong generalization and linear relationship withing features which are processed and optimized by feature engineering. Ratio-based features and log transformation was helpful to get realistic results by reducing skewness and identifying documentation density. Target Leakage was avoided by controlled Gaussian noise and feature removal strategy and that lead to get an accurate result.

When it comes to confusion matrix it clearly showed that moderate false negative and false positive rates which provides a better output regarding the model accuracy and ROC curve which represented good class discrimination capability and a steep curve.

6. Real-World Deployment

The proposed solution to conduct a successful analysis using EHR data and this EHR quality scorecard system can use as a tool to get validated and quality data before conducting the analysis and this machine learning model can be used to integrate the with hospital analytics systems, researches and scientist can see the data quality score and this could be beneficial to lead a unbiased researches and right clinical insights. There is a key risk when relying on automated data scores and because of the human review is when making decisions.

6.1 Potential Stakeholders

This machine learning model is designed to research institution hospital systems and government institutions and for academic bodies who are seeking to make a healthcare decisions and for the healthcare analytics teams. Model's nature makes it accessible and focused on research efficient low 5 biased data and score card shows the quality level of data then users can rely on EHR data. Solution can be introduced as a low-cost or open-source solution as accessible for the stakeholders. Furthermore, to extend its reach while enhancing the scalability and cost efficiency, cloud-based deployment can be implemented.

6.2 Benefits

This model support to identify research-ready electronic health records by using those data hospitals can make rules and regulations and understand which type of data can make an impact this could support to improve clinical studies overall this system supports to data utilization, healthcare research planning and operational efficiency in healthcare sector

6.3 Risks

Even though this system has a more advantages there's a certain risk and limitations. The research-ready label was constructed synthetically which is not fully represent real-world complexity. In different healthcare institutions has different data distributions because of that prediction could be affected by that. This is recommended to use as a decision support tool rather than expert evaluation.

7. Conclusion

This project demonstrates the importance of utilizing quality EHR data and demonstrates a solution to enhance data before applying machine learning and analytics. By introducing the EHR Data Quality Scorecard, it enables the early detection of records which are not reliable, upgrade fairness, and improve reproducibility in healthcare research. To summarized, this approach highlights how quality data are being structured using engineering practices and intelligent data assessment can be a huge impact on enhance the prevailing standard of real-world healthcare systems.

8. Future Implementations

1. Validating the model using expert-labelled datasets which are available in the real world as it ensures the generalization of real data.
2. Incorporating structured and unstructured clinical data additionally such as results from laboratories, medication history (structured data), and clinical notes (unstructured data).
3. Implementing interpretability methods such as SHAP (Shapley Additive Explanations) to increase transparency
4. Training the model with advanced machine learning algorithms such as Gradient Boosting, or deep learning architectures (neural networks).
5. Developing real-time decision-support system ensembles with hospital information systems (HIS) or EHR platforms

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Appendix

- **Data Source -** https://github.com/syntheticheath/synthea
- **GitHub Repository** – https://github.com/Thilini-Piyumika/Electronic-Health-Record-Quality-Scorecard (included: Collab Notebook, Aggregated Dataset and Data Tables)

Aggregated Dataset

	A	B	C	D	E	F	G	H	I
	Id	number_of_visits	number_of_observations	number_of_conditions	number_of_medications	number_of_procedures	number_of_immunizations	number_of_imaging	number_of_allergies
	3b2332c2-fab9-0b43-dc7a-15fb1f29411a	20	200	10	3	24	27	1	2
	ac0ab69a-b9bf-82b1-120c-65c417a442eb	21	235	13	2	11	33	0	0
	455ee5d2-390e-3315-8f83-ba9d64fc2bff	20	214	12	4	15	33	1	0
	e619d586-c412-226d-c113-77214b7193e6	15	220	18	5	80	9	1	0
	4055a7e2-bea9-b799-f9f0-3cb8f418971d	27	265	17	6	26	36	1	0
	c490ed81-717d-d6a0-397a-3a3e0636a12b	20	247	22	7	79	8	2	0
	c7e39698-dc0c-52d5-e1c1-41941c0371fa	66	474	38	18	162	13	4	0
	e9b51a44-98d0-d549-778a-b45d1aa9f4d5	35	408	28	29	97	10	2	7
	eda8bb59-4172-5cc3-8b38-881856d1f501	69	196	22	10	230	7	2	0
	1294e3f3-6cbd-ef3f-7265-27ff53a57ce8	81	416	39	29	150	16	3	0
	ecf9d606-18c8-7b7d-31fd-04147ca41201	88	177	28	11	214	9	2	8
	8c1974ab-350c-1d4e-973a-ff098e6c13a7	49	900	37	53	197	9	1	0
	6ba3d2d2-c3f0-13e4-679d-beeeac2b51e5	17	206	18	6	70	12	4	0
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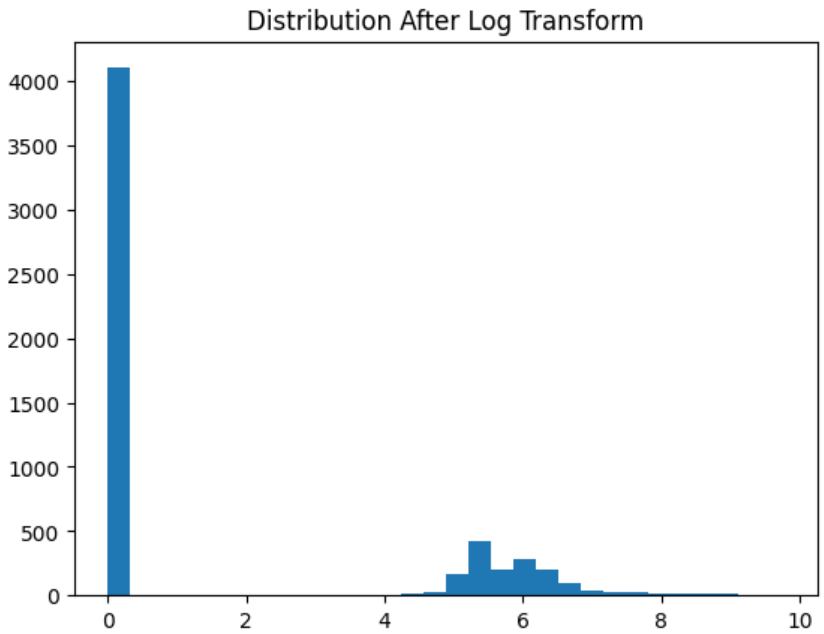
	0b44e59a-231b-b496-3f36-53a582fd288d	57	376	19	18	183	11	3	0
	34c5dd25-7503-39c0-4dd0-11eefb961e61	27	253	14	8	40	36	0	0
	b771fe6c-5cc6-cdc8-7743-ac4ba9155b83	30	398	31	16	107	15	1	0
	5a0c0921-b416-8852-b044-e1a4affd632d	29	236	23	11	129	17	3	0
	b44da956-a099-a539-69f2-7453a01b1f68	21	387	25	11	71	15	1	0
	de950226-157d-ddc3-3c0b-6762fbea9f3c	11	100	5	1	22	4	0	0
	9109a701-12aa-1d92-02f4-87b35fbf3800	17	271	15	5	39	10	0	2
	5b5cf06e-4de1-e99e-3d61-3e45ba4df180	31	406	31	11	91	13	4	0
	164ceae3-2d92-56eb-4374-34da12d701df	75	345	39	5	117	12	4	0
	42750e78-10bd-da68-a341-90eec978414b	29	437	37	3	58	14	0	0
	409e47cb-43dd-1ea5-7f27-b4b3e64e6655	63	376	36	41	98	13	3	0
	73438c45-8781-8f59-b0fb-6f03a46bf0ae	13	205	17	0	47	12	1	0
	5d860b04-1574-045e-3735-075032945d17	56	692	51	91	160	13	5	0
	4a84aed1-8cdd-fd98-da35-05f87d62e198	74	422	42	38	219	12	3	2
	586539c4-5844-7ade-2223-d9c86f80bea2	53	259	29	13	145	18	4	11
	10513a4c-0539-8781-b048-f2287b7908ea	52	1452	69	59	247	15	7	0
	658261da-b3ee-2ee1-10f7-669b6ef028fc	29	267	25	5	47	11	1	0
	6e37751f-c685-b931-c895-bdcfee9ccfa8	110	666	31	50	314	21	40	1
	caa78af5-3487-f7b0-925d-7c285663a73c	53	273	27	7	189	9	1	0
	171d1c73-6056-4a27-929d-e55d881f97c9	25	233	14	7	83	13	2	0
	f3a21847-0fab-4af7-bf19-85ed70eb4687	21	212	11	5	26	32	2	0
	8ab1283d-1cff-8f66-3375-758a6e49d75a	24	243	20	6	112	17	3	0
	82bb7473-3209-3232-4cfe-5cca89acdb35	58	413	32	28	115	17	4	8

patients | encounters | observations | conditions | medications | procedures | immunizations | imaging_studies | allergies

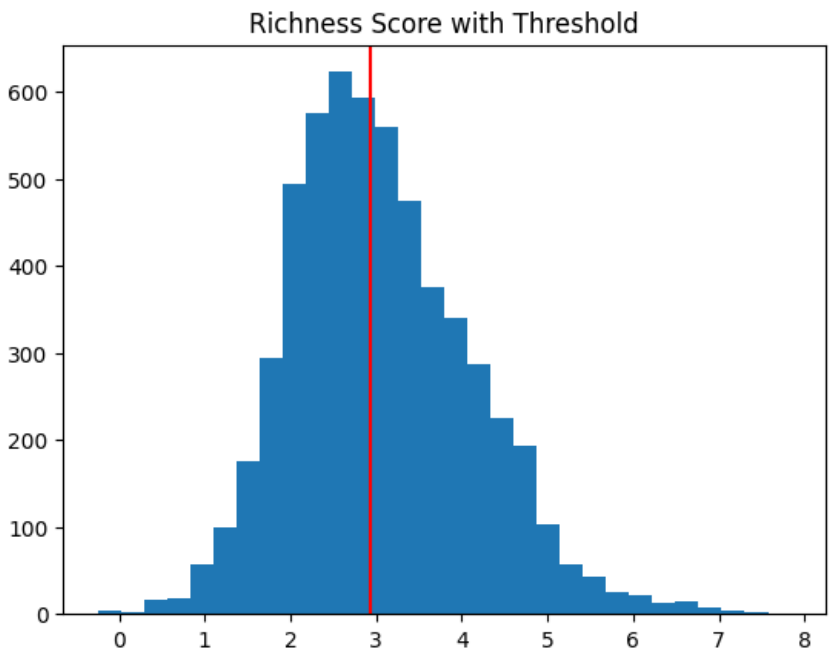
Formula used in Excel sheet to Aggregate data tables

=COUNTIF(observations!B:B,A2)

Distribution After Log Transformation



Richness Score Distribution



Calculations

1. Log Transformations

Formula: $X' = \log(1 + X)$

$X = 99$

$1 + 99 = 100$

$\ln(100) = 4.605$

Final Answer: $99 \rightarrow 4.605$

2. Standardization

Formula: $Z = (X - \mu) / SD$

$X = 10,$

$\mu = 8, SD = 2$

$10 - 8 = 2$

$2 / 2 = 1$

Final Z-score = 1

3. L2 Regularization

Loss = Original Loss + $\lambda \Sigma w^2$

Original Loss = 0.40

Weights- $w1 = 2, w2 = 3$

$2^2 = 4, 3^2 = 9$

Sum = 4 + 9 = 13

$\lambda = 0.1 \rightarrow 0.1 \times 13 = 1.3$

$0.40 + 1.3 = 1.7$

Final Regularized Loss = 1.7

4. Gaussian Noise Addition

$R' = R + \varepsilon$

$R = 5.2,$

$\varepsilon = -0.3$

$5.2 - 0.3 = 4.9$

New Value = 4.9

6. **Accuracy Calculation**

$$\text{Accuracy} = (TP + TN) / \text{Total}$$

$$TP = 655,\ TN = 738,\ FP = 118,\ FN = 200\ \text{Total} = 1711$$

$$655 + 738 = 1393$$

$$1393 / 1711 = 0.814$$

$$\text{Accuracy} = 81.4\%$$

7. **Precision**

$$\text{Precision} = TP / (TP + FP)$$

$$= 655 / (655 + 118)$$

$$= 655 / 773$$

$$= 0.847$$

$$\text{Precision} = 84.7\%$$

8. **Recall**

$$\text{Recall} = TP / (TP + FN)$$

$$= 655 / (655 + 200)$$

$$= 655 / 855$$

$$= 0.766$$

$$\text{Recall} = 76.6\%$$

9. **F1 Score**

$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

$$= 2 \times (0.847 \times 0.766) / (0.847 + 0.766)$$

$$0.847 \times 0.766 = 0.649$$

$$0.847 + 0.766 = 1.613$$

$$2 \times 0.649 = 1.298$$

$$1.298 / 1.613 = 0.804$$

$$\text{F1 Score} = 80.4\%$$

10. Cross-Validation Mean

Scores = [0.80, 0.82, 0.79, 0.81, 0.80]

Sum = 4.02

4.02 / 5 = 0.804

Mean = 0.804

11. Standard Deviation

Mean = 0.804

Differences:

0.80 - 0.804 = -0.004

0.82 - 0.804 = 0.016

0.79 - 0.804 = -0.014

0.81 - 0.804 = 0.006

0.80 - 0.804 = -0.004

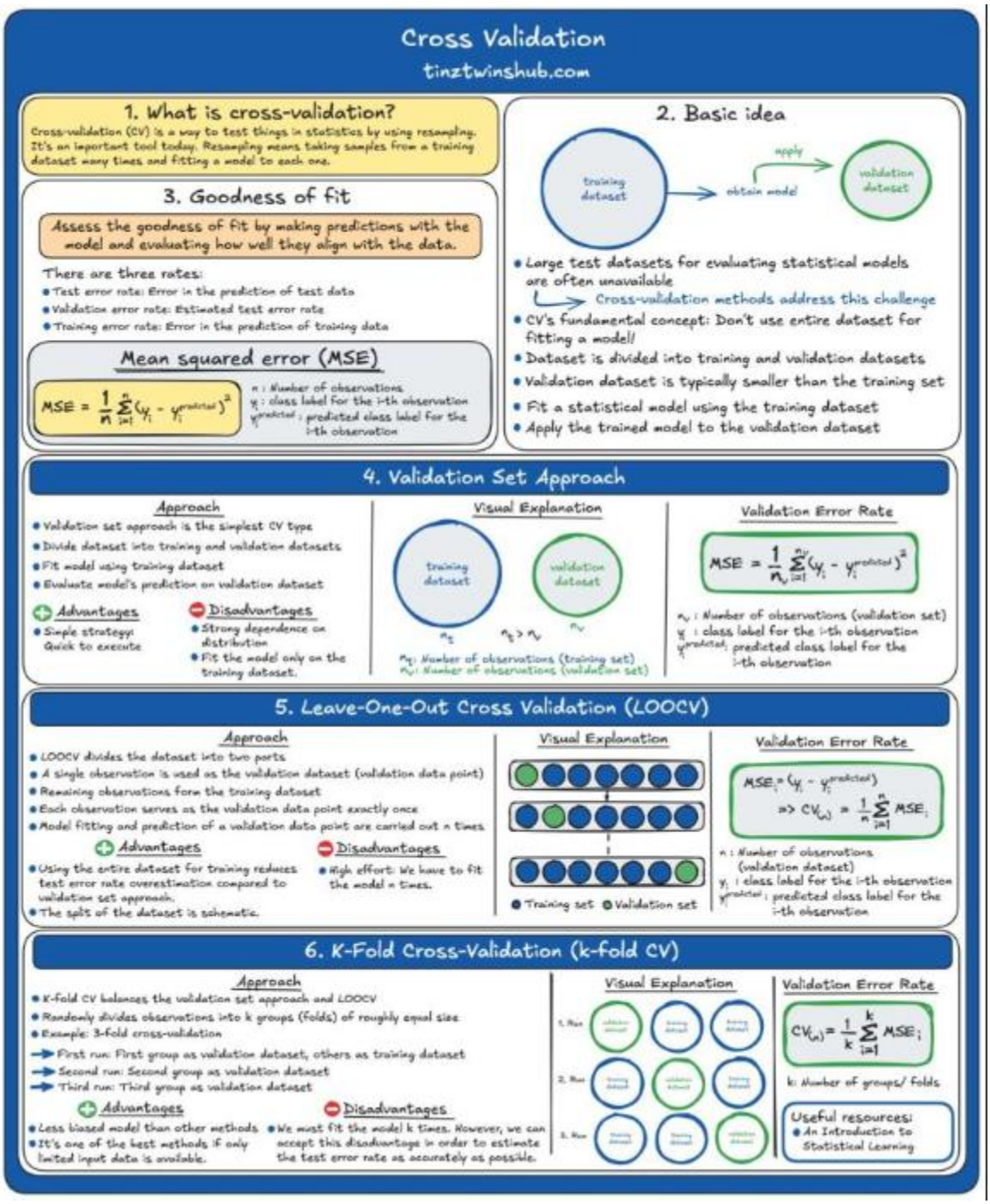
Squared Differences: 0.000016, 0.000256, 0.000196, 0.000036, 0.000016

Sum of squares = 0.00052

Variance = 0.00052 / 5 = 0.000104

Standard Deviation = $\sqrt{0.000104} = 0.0102$ $\sigma \approx 0.01$

LinkedIn Article About Cross-Validation



- End of the report -